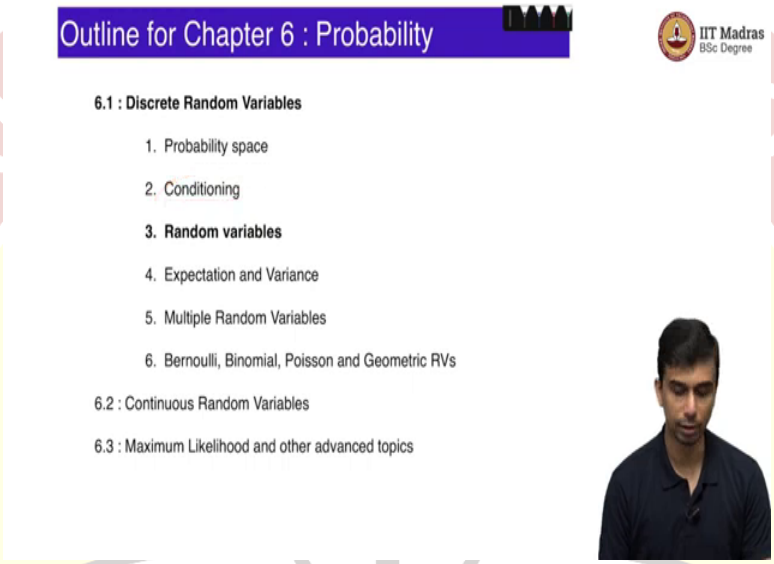



# IIT Madras

## ONLINE DEGREE

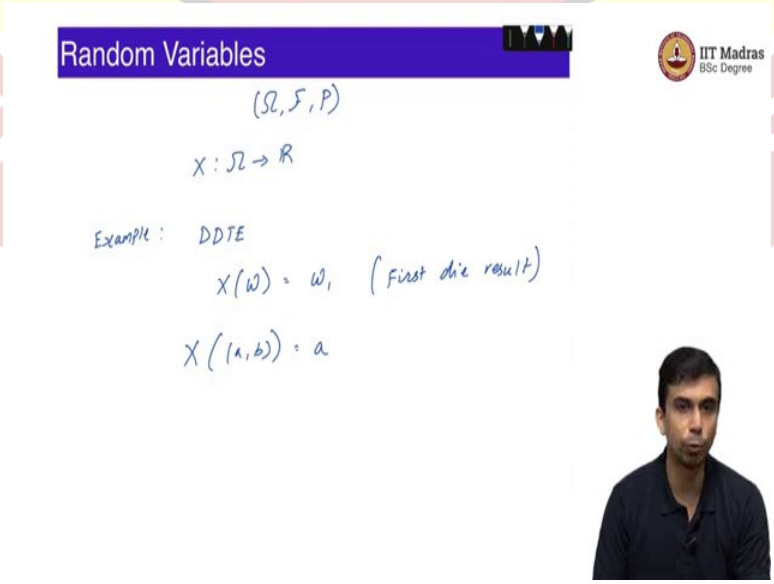
**Machine Learning Foundations**
**Professor. Harish Guruprasad Ramaswamy**
**Department of Computer Sciences & Engineering**
**Indian Institute of Technology, Madras**
**Random variables**

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Hello, everyone. Now that you have seen the basics of probability space and conditioning, we can move to the next part of the chapter which is Random Variables.

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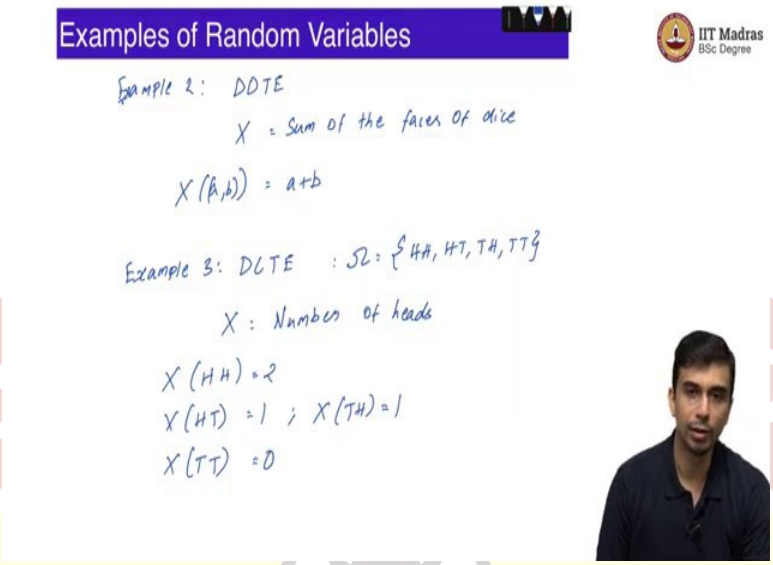
We have already introduced the sample space, the sigma algebra, the collection of events and the probability measure. In this context of an experiment, we can define a random variable. A random variable, mathematically speaking, is nothing but a function from the sample space to the reals. So, that is, a random variable is just a function. If the random variable can take as output as the range of the random variable is only a discrete set, then it is called a discrete random variable. In particular, the range is finite, then there are only a finite number of values we can take, then this is a discrete random variable.

In particular, for the examples that we have seen till now their omega, the sample space itself is finite. Clearly, the set of outputs values that this random variable can take is also going to be finite. So, all such random variable are going to be discrete random variables. That is why you call discrete random variables, because they can take only a discrete set of values. And so it is easier for us to analyze such random variables.

So, what are some, why is random variable defined this way and does it make intuitive sense? And so let us see some examples to go with this first. Let us go with the first example in the double dice throw experiment. Recall, the double dice throw experiment, your sample space was of the form a comma b, where a and b are in the range of 1 to 6. The cardinality of the sample space was 36.

So, let us say the x of omega, omega is an element in the sample space is omega 1 or rather the first die result. That is a random variable. How does it make sense? For example, if the sample space element, let us say, is a comma b, that is equal to a. So, if you roll the dice and it is you get a 4 comma 3, the random variable x corresponds to 4, which is the first dice. So, that is just very simple random variable.

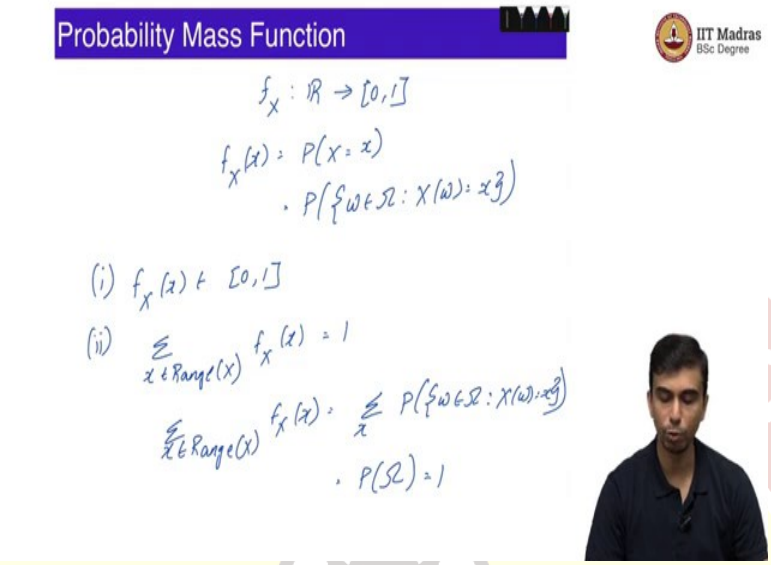
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Let us go with one, give one more example. Example 2. Let us go the same double dice throw experiment. Let us have x is the sum of the faces of the dice, which means that x of a comma b is simply equal to a plus b. This is also a simple random variable and it is an intuitive random variable.

Let us go, let us give one more example. Let us take the double coin toss experiment. Let us say x here responds to the number of heads that you see. Here in the double coin toss experiment your omega is head heads, heads tails, tails heads and tails, tails. So, in this case x of HH would be equal to 2, x of HT would be equal to 1, x of TH would also be equal to 1 and x of TT would be equal to 0. So, this is another simple example of random variables. Random variables are a powerful tool. And there will be, we will analyze random variables with several attributes.

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One of the more basic attributes that we will use is the probability mass function, which is typically for a random variable x is denoted by small f, and it is often put a subscript indicating the random variable fx. What is fx? So, fx is a mapping from the range of the random variables to the reals again, R positive reals. This case it is just 0 to 1. So, the probability mass function is a function which takes in a value and outputs a number of 0 to 1. What is the meaning of fx of x?

Well, fx of x is simply the probability that the random variable x takes the value x. So, strictly speaking, this is not defined, because probability measure, the probability measure takes only events as inputs. So, x equal to x is not, it is not clear what the event is. We will expand what that means. It is the set of all omega in omega such that x of omega is equal to x. So, we will use, this denotes small omega or lowercase omega, this is capital omega and this is capital X and this is small x, but I will call just x and omega. From context it should be clear whether I am referring to the lowercase omega or uppercase omega and similarly for x.

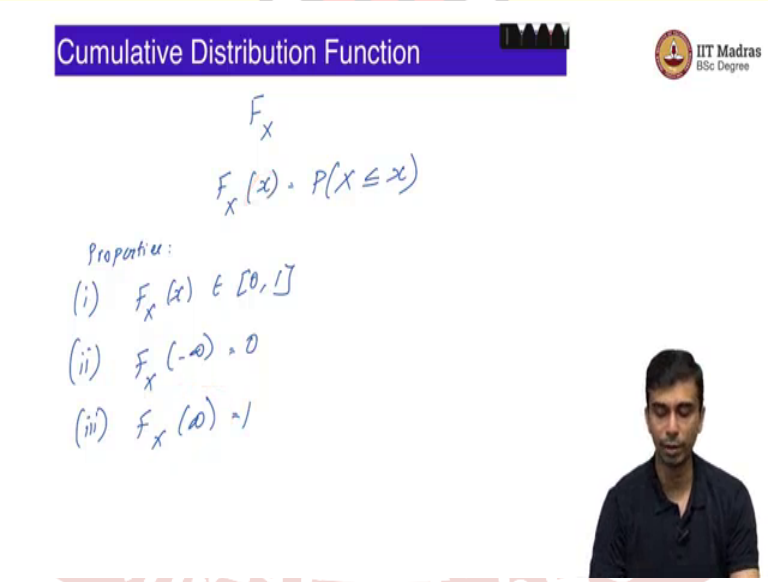
This is an event. Why is that, because, this first of all, it is a subset of the sample space omega. It consists of all the elements in the sample space, such that your X of omega equal to x. And this is a valid event because all subsets of the sample space are right now valid events. And you can compute the probability of such an event. So, that is the probability mass function.

What are some important properties of the probability mass function? Well, first of all, the first property is that the fx of x belongs to 0 to 1. It cannot go below 0, it cannot exceed 1. Why is that

because the probability that is, as a consequence of the probability axioms. That is the first property. And the second property is also comes as a consequence of the axioms of probability, which is that if you sum over all x, in the range of the random variable X, recall range of a function is simply the set of values, the output can take of that function can take x, sum over x in range of X fx of x is equal to 1. So, this is the second property. And this also is a consequence of the probability axioms.

Why is that? Well, fx of sum over x in the range of X, fx of x is simply sum over x probability of the set of omega in omega such that x of omega equal to x. Note that each of these events set up omega and omega set of x, omega equal to x for different x's are disjoint. And their union contributes, their union is equal to the entire sample space, which means that this is exactly equal to the probability of omega which is equal to 1. These are the two main properties of the probability mass function. That is its values lies between 0 to 1. And if you sum over all its nonzero values, it is going to be equal to 1. So, that is the probability mass function.

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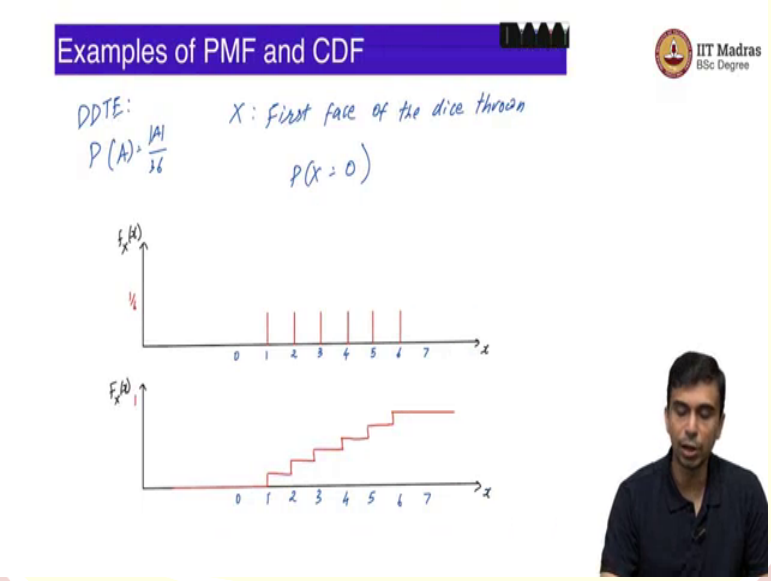



There is another attribute that is to the random variable which is often studied, which is the cumulative distribution function and it is often denoted by capital F and a subscript of x. What is capital F subscript x of, let us say, some x, which is probability of this random variable X taking values less than or equal to that small x. So, this is the cumulative distribution function.

So, what are the properties of a cumulative distribution function? Well, the properties, the first property is that once again fx of x lies between 0 to 1. It cannot take just this once again as a consequence of the axioms of probability. Probability measure takes only values between 0 to 1.

The second is slightly also interesting. This is also a consequence of the definition of the cumulative distribution function are the CDF, which is Fx of minus infinity is equal to 0 and Fx of infinity is equal to 1. Why is that, because I mean we are, we do not, we can simplify this even further, which is for, if the small x is let us say minus infinity, probability of x less than equal to x is clearly 0. And if your small x is infinity, probability of x is equal to x is clearly equal to 1. So, these are other properties for the CDF.

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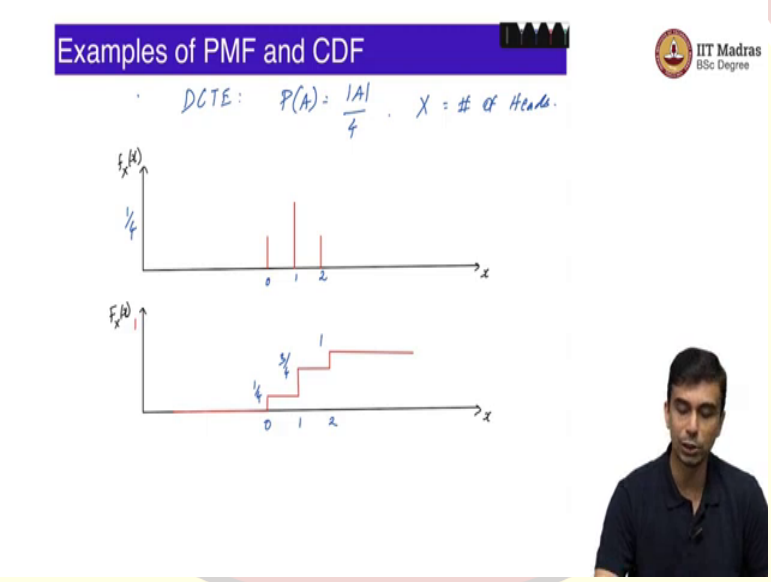

So, let us see some example random variables and their probability mass function and CDF. Let us take the first example random variable that we took, where x in the double dice throw experiment with the probability measure being the counting measure, that is probability of A is simply equal to cardinality of A by 36. So, this is the counting probability measure. So, in this case, let us say, x is the single, first face of the dice throw, of the dice. So, what is the probability that let us say x is equal to say 0, what is that? Well, the answer is 0. So, that is why you can see that.

When you plot the probability mass function, here you get 0. It is actually only non-zero for these values 1, 2, 3, 4, 5 and 6 only at those points is the probability mass function non-zero and

it takes the value 1 by 6, and everywhere else it is 0, that is the probability mass function. You can do the same thing and plot the CDF or the cumulative distribution function also, which is you can see that it is going to be 0 for anywhere below 1.

So, fx of 0, capital Fx of 0.99 is going to be equal to 0, because the probability of x is equal to 0.99 is equal to 0. The moment you hit 1, though, that is probability of capital X, the first phase being less than or equal to 1 is exactly equal to 1 by 6. When you go to 2, probability of x less than or equal to 2, it becomes 2 by 6, between 1 and 2, it stays constant at 1 by 6. So, this is how the CDF or the cumulative distribution function of this particular random variable looks like.

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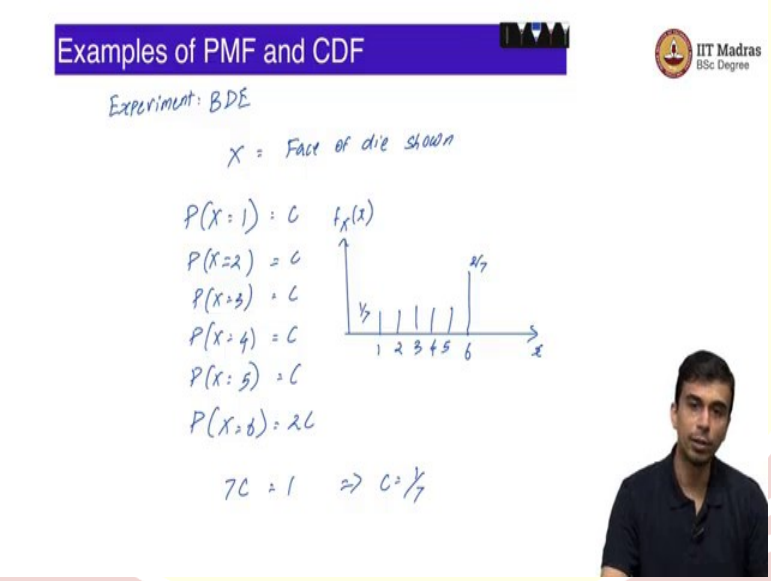

Let us take another example. Let us take the double coin toss experiment with the once again the equal probability that is counting probability measure, that is probability of A is simply equal to cardinality of A by 4, because the sample space consists of four elements. So, this is the probability of A simply cardinality of A by 4. The random variable that we are going to, we interested in this x is equal to the number of heads.

What is the probability that you will see 0 heads? Well, that is equal to the probability of the event TT, which is equal to 1 by 4, sorry, 1 by 6 is 1 by 4, what is the probability that you will see one head, that is exactly the probability of the event HT comma TH which is 2 by 4, which is 1 by 2.

And what is the probability that x is equal to 2, which is the probability of number of heads being 2, which is exactly equal to the property of the event, the singleton event HH, which is also equal to 1 by 4. And you can do that. So, the probability mass function of this random variable looks like this. It is 0 everywhere else outside 0, 1 and 2, at 0, it takes the value 1 by 4, at 1, it takes the value 1 by 2, and at 2 it takes the value 1 by 4. You can get the cumulative distribution function of this random variable also. It us pretty easy.

So, until 0, it is the CDF stays at 0. At 0, it climbs up to 1 by 4, at 1 it climbs up to 1 by 4 plus 1 by 2, which is 3 by 4, at 2 it becomes 1, and it stays at 1 all the way till infinity. Because, so that is the reason I can see that at the leftmost end of the curve, your CDF is going to be always 0 and the rightmost end it is always going to be equal to 1. So that is the, you can see the properties of the CDF and the PMF manifesting here in these examples.

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Let us do one slightly more complicated experiment, which is, let us say you are a gambler and you are, you have a dice which is biased. The way you have constructed the dice is that with the dice can throw 6 twice as likely as the other faces. All the other faces are equally likely, but the face 6 is let us say weighted or let us say magnetic or something like that so that the probability of 6 is twice as much as the other face. Now, we want to ask the question, what is the probability that the face is let us say 3?

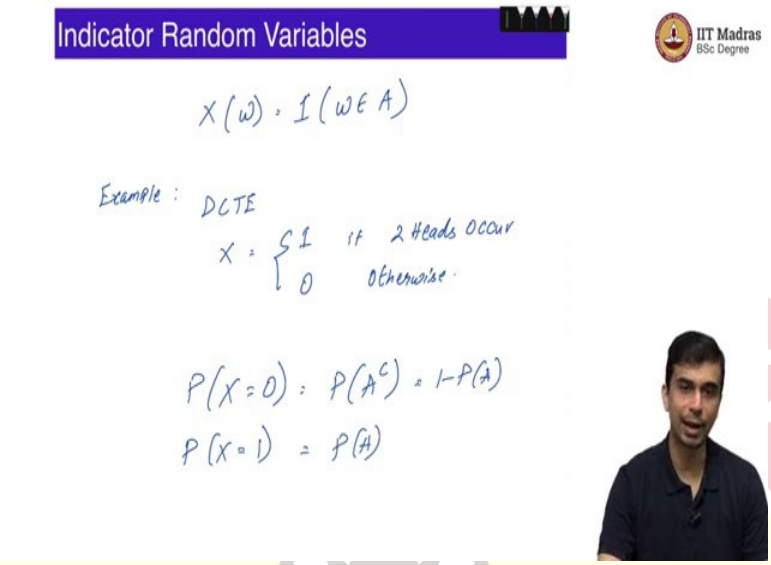
So, now this is, let us call this experiment as the biased dice experiment. This is just a single dice, biased dice experiment. Now the way we have constructed it, we have constructed this bias dice. The question I am asking is, let us say, x is the random variable of that incident is the result the face of the dice shown. What is the probability that x is equal to let us say 1? What is that?

We have not explicitly given the answer here. The only thing we have said is that all the faces 1 to 5 are equally likely and 6 is twice as likely as that. So, let us call probability of x equal to 1 equal to let us say C. Similarly, you have probability of x equal to 2, 3, 4, 5 all of that are equal to C. Probability of x equal to 5 is equal to C. So, your probability mass function looks like C, C, C, C, C, C. What is the probability of x equal to 6? We know that is 2C.

From the properties of the probability mass function, if you sum over all of these values, you get 1, which means that 7C should be equal to 1. So, which implies C should be equal to 1 by 7. So, the probability of this particular dice throwing 1 is equal to 1 by 7, probability of this particular dice throwing 2 is 2 by 7, 1 by 7, and so on. So, 1 to 5, the probability of the dice showing that number is 1 by 7, probability of that dice showing the result 6 is 2 by 7. So, this particular random variable has this PMF given like this, looks like this.

So, 1, 2, 3, 4, 5 these values are 1 by 7. Here, it is 2 by 7. So, this is the PMF of this particular element. So, this is some examples of PMF and CDF and how you can explore the properties of the PMF to get concrete answers. Even though the English version was not completed, I mean, not, did not give you the full answer immediately. It just said relatively 6 is twice as likely as the other faces, you can use the property of the PMF to get the exact probability of this particular dice rolling a 3 or a 4 or a 6.

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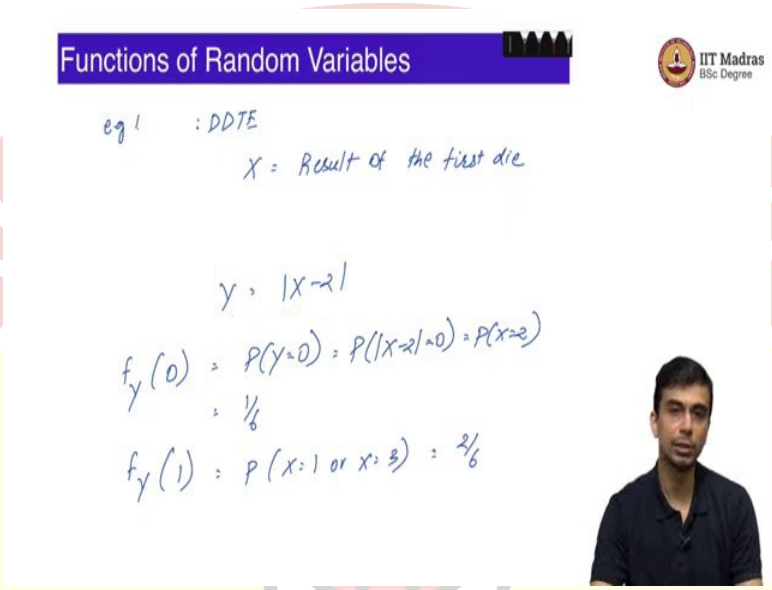

So, there is one other special class of random variables which are, which occur very often, especially in analyzing other random variables, which are indicator random variables. So, indicator random variables are special because they take only two particular values, which is either 0 or 1. Indicator random variable x of omega is simply equal to indicator of omega belonging to the set A. This A is an event. And this random variable simply is equal to 1 if that event happens and 0 otherwise. This is called an indicator random variable.

An example would be, let us say, let us take the double coin toss experiment. I can have a random variable x is equal to 1 if the, if you have two heads, two heads occur and 0 otherwise. This is an example of indicator random variable because it can take only two possible values, 1 or 0. So, what is the probability that x is equal to 0?

Well, this is exactly equal to the probability that let us say x corresponds to the indicator random variable for the event A. So, this is exactly the probability of A complement, which is 1 minus probability of A. What is the probability of x is equal to 1, which is exactly equal to probability of A. So, A is the, we can have an indicator random variable corresponding to an event A, which is probability, the probability of x equal to 0 is 1 minus probability of A and probability of x equal to 1 is simply probability of A.

So, in this particular example, A is the event that two heads occur that is the event. So, these indicator random variables are very useful for analyzing, for building up more complicated random variables. We will see that in a later chapter.

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So, moving on to another useful tool which is, we can have functions of random variables. The same way you can have functions of real valued constants or variables, you can have functions of random variables also. For example, let us say you have the function, let us take the simple experiment. Let us take example 1, which is let us say the double dice throw experiment. X is the result of the first dice.

So, in this case, you can have a function of this random variable. You can define y to be equal to let us say x square or you can define y to be equal to absolute value of x minus 2. You can have these are all functions of the random variable x. And they themselves are random variables. For example, let us take y equal to absolute value of x minus 2. Let us take this example. What is the, y itself is a random variable.

What is the probability mass function of this random variable y. what is fy of let us say 0 which is probability of y is equal to 0, which is exactly equal to the probability of absolute value of x minus 2 is equal to 0, which is equal to probability of x equal to 2. Let us say if it is the counting probability measure, this is exactly equal to 1 by 6.

What is fy of 1 is probability of y equal to 1 which means that probably the absolute value of x minus 2 is equal to 1 which is equal to the probability of x equal to 1 or x is equal to 3. This is equal to 2 by 6, and so on. So, you can compute the probability mass function of this derived random variable in terms of the, by using the property mass function of the original random variable. So, this is another tool that we will often use for analyzing random variables. So, with that, we will wrap up this lecture and we will move on to the next topic.

